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Multimodal Chatbot for Plant Disease Recognition

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December 7, 2024

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- Agriculture is a critical sector for global food security. Plant diseases can significantly impact crop yield and quality.
- Farmers often struggle to identify diseases promptly due to a lack of access to expert knowledge.
- A chatbot can bridge this gap by providing instant disease recognition and treatment suggestions.



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To develope an AI driven chatbot that can accurately recognise plant disease from the leaves of the plant and answers the user's query through the chatbot.



OBJECTIVES

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The major objectives of the project are as follows:-

- Design a chatbot capable of handling diverse queries and scenarios, making it adaptable for a wide range of plant species.
- Enable the chatbot to understand and respond to users' query.
- Ensure the chatbot maintains context throughout conversations, allowing it to provide coherent and relevant responses, even during extended interactions.
- Create a user-friendly interface that is easily accessible to individuals across different demographics, including those with limited technological proficiency.



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- To provide a user-friendly and efficient solution for plant disease identification, decreasing reliance on expert analysis and encouraging timely intervention.
- To use multimodal AI models to process image inputs and generate descriptive textual outputs while providing clear and actionable recommendations.
- To increase agricultural productivity by properly recognizing plant illnesses and offering timely remedies using a chatbot interface.
- To improve the efficiency of plant disease detection by incorporating cutting-edge AI technologies like as MobileNetV2 and RAG models.
- Create a user-friendly and interactive solution with Streamlit for farmers and agricultural stakeholders.



LITERATURE SURVEY

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It reviews various studies focused on applying machine learning and deep learning techniques for plant disease detection, highlighting the datasets, methods, and challenges involved. Many of the studies utilize commonly available datasets such as the PlantVillage dataset[1], along with specialized datasets like the Indian Medicinal Leaves and Potato Leaf Disease datasets.[2] The methods employed range from Convolutional Neural Networks (CNNs) and Transfer Learning to advanced architectures like AELGNet, MDSCIRNet, and CoffeeNet. These models often achieve high accuracy rates, with some surpassing 90 percentage[3], showcasing their potential for real-time disease detection.



BASE PAPER DISCUSSION

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The base paper been choosen for this project is **Deep learning models for plant disease detection and diagnosis.**

- Approach: Recognising disease through Convolutional Neural Networks
- Models used: CNNs.
- Results: 91.50 percentage accuracy, indicating that they might be used as an advisory tool in real cultivation situations.
- Research Gap: Increased database size, training and test set overlap, lower performance with diverse data, real-world identification complexity, shading, and irrelevant objects.



PROPOSAL

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I recommend using MobileNetV2 for efficient plant disease detection on mobile and edge devices, offering high accuracy with low computational overhead. By integrating Retrieval-Augmented Generation (RAG), the system will enhance predictions with context-aware, real-time recommendations, providing a scalable and practical solution for plant disease management in agriculture.

SOME IMPLEMENTATION ON PROPOSAL

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This is a try to develop and fine-tune MobileNetV2 model





TIMELINE

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- MobileNetV2 model development and Fine-tuning
- RAG integration and testing
- Deployment and UI
- Testing and Documentation



CONCLUSION AND FUTURE WORK

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- This project highlights the potential of deep learning in agricultural diagnostics, providing an automated solution for disease detection that can aid in crop health monitoring.
- Implementing MoblieNetv2 and RAG LLM model on diverse dataset.



REFERENCES

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